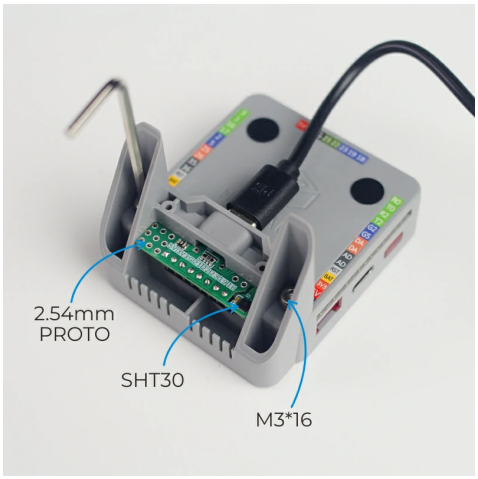
Base BTC

SKU:A011/A011-B











Description

Base BTC is an environmental data measurement base with a built-in SHT30 temperature and humidity sensor, capable of precise measurements. It adopts a female header plug-in design, connecting and communicating with the core controller through the Base Bottom pin header. It is suitable for building personal desktop environmental data monitoring and display systems.

Features

- Built-in SHT30 temperature and humidity sensor
- High measurement accuracy
- I2C communication interface

Includes

- 1 x Base BTC
- 1 x USB Type-C Cable (1m)
- 2 x M3 * 16 Screws
- 1 x Hex Key L-Shape 2.5mm (For M3 Screw)

Applications

- Indoor environment monitoring system
- Smart home linkage devices
- Educational experimental platform
- Small weather station

Specifications

Specification	Parameter
Maximum Temperature Range	-40 ~ 120 ℃
Highest Measurement Accuracy	0 ~ 60 ℃/±0.2℃
Humidity Range / Error	10 ~ 90 % RH / ±2%
Product Weight	9.2g
Gross Weight	60.0g
Product Size	55.0 x 47.0 x 23.0mm
Package Size	95.0 x 66.0 x 26.0mm

Learn

Charging Instruction

When used with first-generation Core series controllers such as Basic:

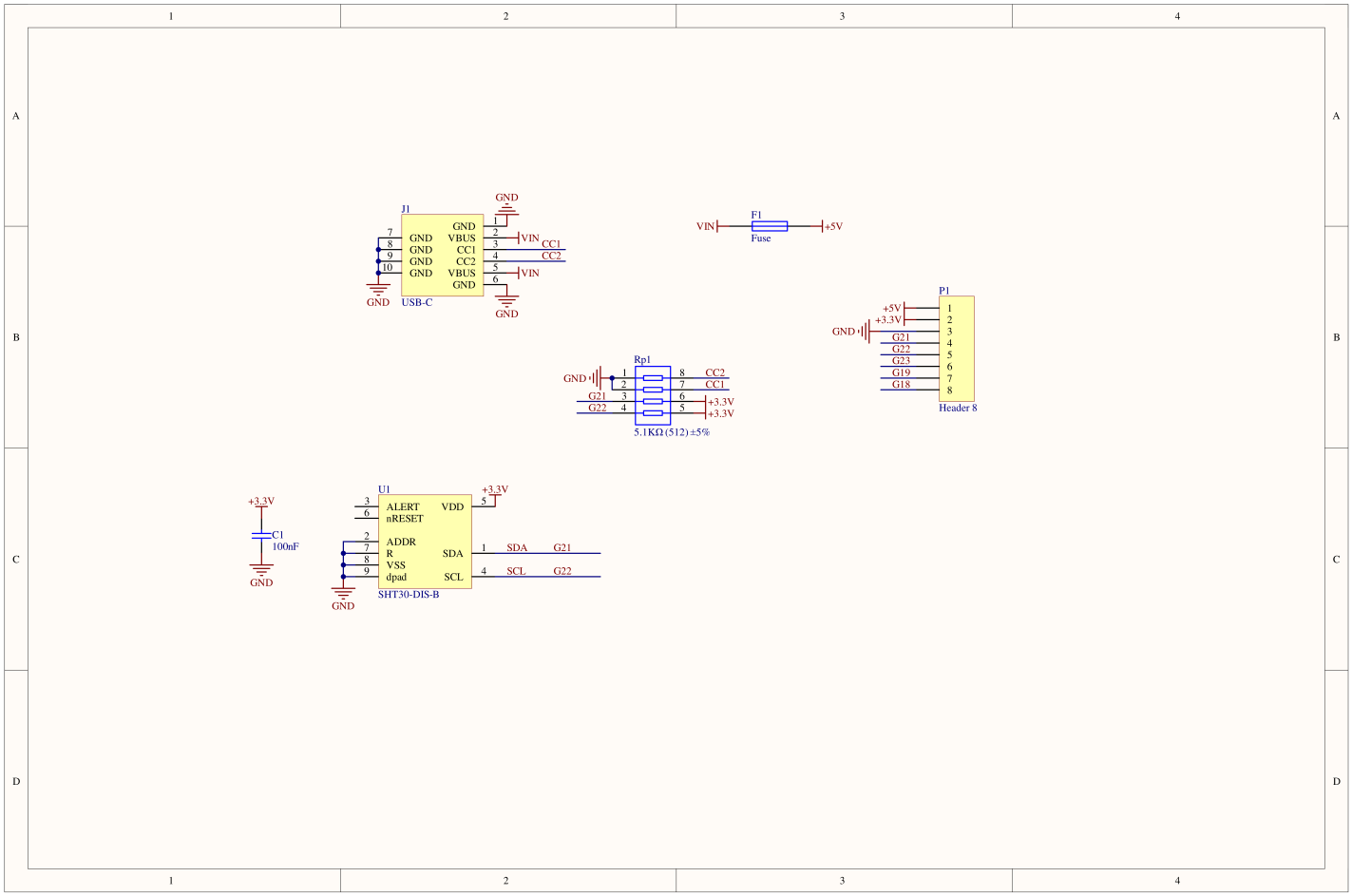
- The USB port of the Base BTC base only supplies power to the host device.
- To charge the lithium battery inside the Base Bottom, please connect the USB port on the host device.

I2C Address Conflict

- Do not connect additional sensors with the same I2C address (0x44) as the SHT30 on the I2C bus, otherwise the device will not work properly due to address conflicts.

Schematics

- Base BTC Schematics PDF



PinMap

SHT30

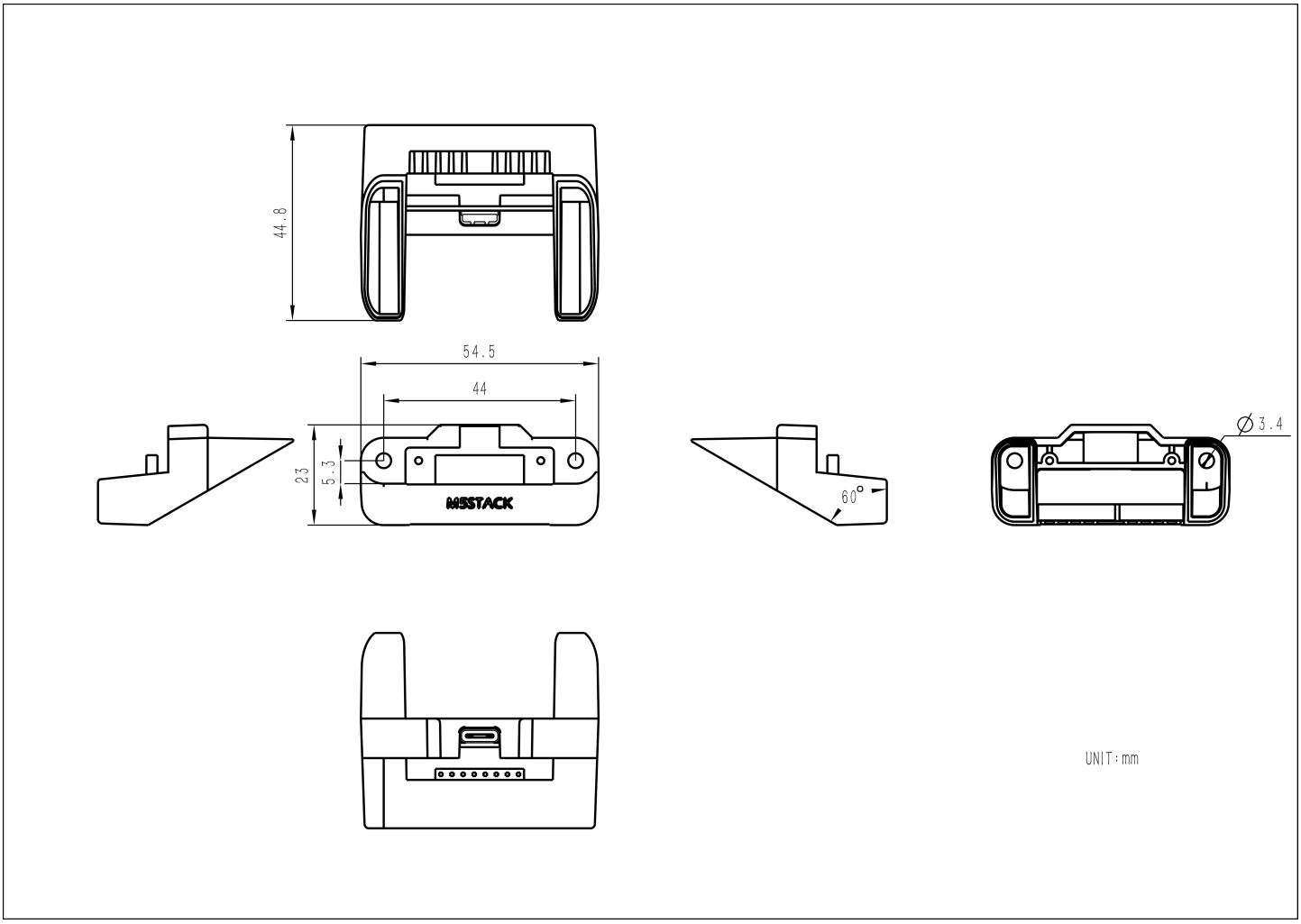
Basic	G22	G21	3V3	GND
SHT30	SCL	SDA	3V3	GND

M5-Bus

PIN	LEFT	RIGHT	PIN
NC	1	2	NC
NC	3	4	NC
NC	5	6	NC
NC	7	8	NC
NC	9	10	NC
NC	11	12	NC
NC	13	14	NC
NC	15	16	NC
I2C_SDA	17	18	I2C_SCL
NC	19	20	NC
NC	21	22	NC
NC	23	24	NC
NC	25	26	NC
NC	27	28	5V
NC	29	30	BAT

Model Size

- Base BTC Model Size PDF



Datasheets

- DHT12 datasheet
- SHT30 datasheet

Softwares

Arduino

- Base BTC SHT30 Example
- Base BTC DHT12 Example

Version Change

Release Date	Product Changes
2020.5	DHT12 changed to SHT30
2017.7	First release